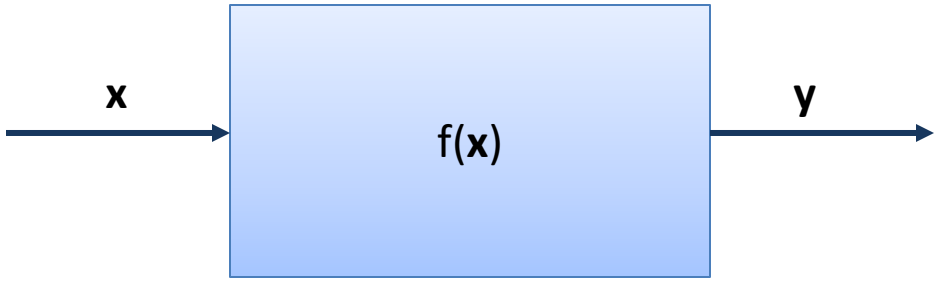
Information Geometry

Than Lwin Aung

Function

Let us consider an **Output Function** f: **x** $\longrightarrow$ **y** which is represented as:

$$\mathbf{y} = f(\mathbf{x})$$



Actually, it is how we generally analyze and design a systems by **Functional Analysis**, such that given Input **x**, what is the Output **y**?

Function in Joint Space

However, let us look at a function f: $\mathbf{x} \longrightarrow \mathbf{y}$ from the perspective of their **Joint Space**, such as ($\mathbf{x}$, $f(\mathbf{x})$) or ($\mathbf{x}$, $\mathbf{y}$).

As we can see, the **Joint Space** of ($\mathbf{x}$, $f(\mathbf{x})$) or ($\mathbf{x}$, $\mathbf{y}$) is actually the subset of their **Product Space**, such that ($\mathbf{x}$, $\mathbf{y}$) $\subseteq$ ($\mathbf{x} \times \mathbf{y}$).

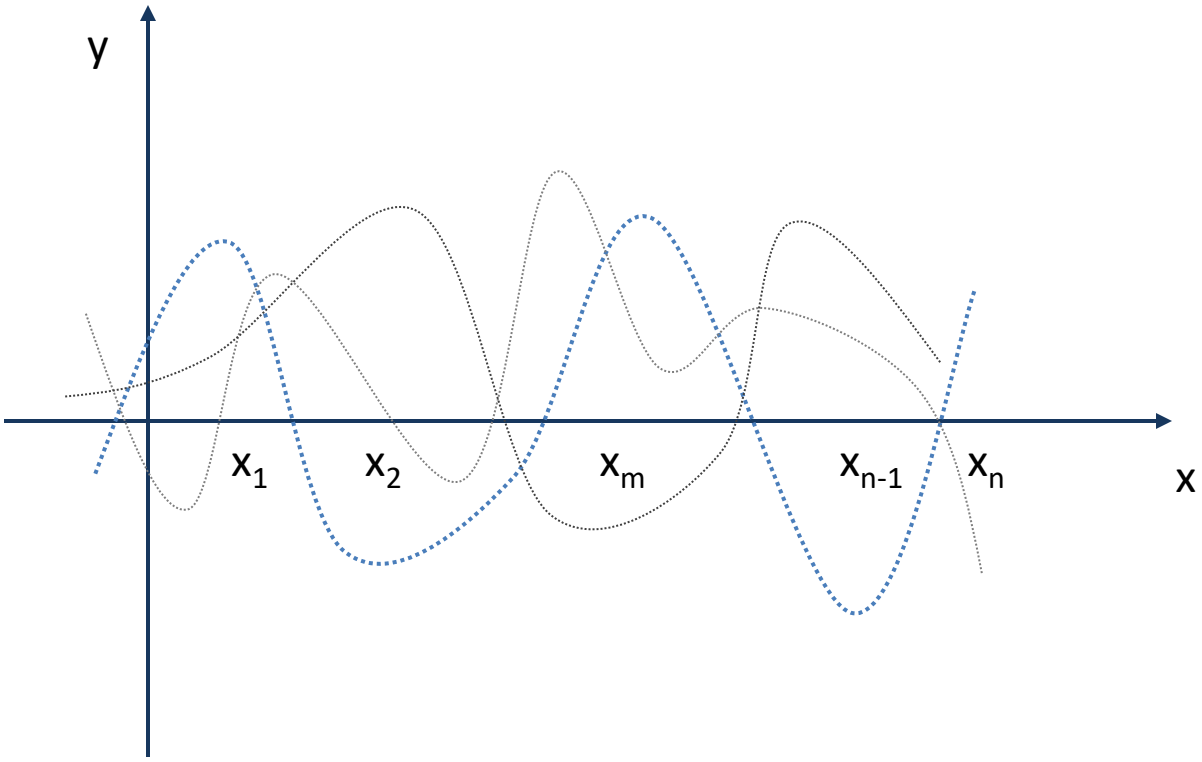
Now, let us ask a question as to whether we can estimate a function f: $\mathbf{x} \longrightarrow \mathbf{y}$, without even knowing what $f(\mathbf{x})$ really is.

To answer this, we need to understand estimating a function f: $\mathbf{x} \longrightarrow \mathbf{y}$ actually means that we are estimating the dependency between ($\mathbf{x}$, $f(\mathbf{x})$) of **Joint Space** in their **Product Space**, such that ($\mathbf{x}$, $\mathbf{y}$) $\subseteq$ ($\mathbf{x} \times \mathbf{y}$).

Function in Joint Space

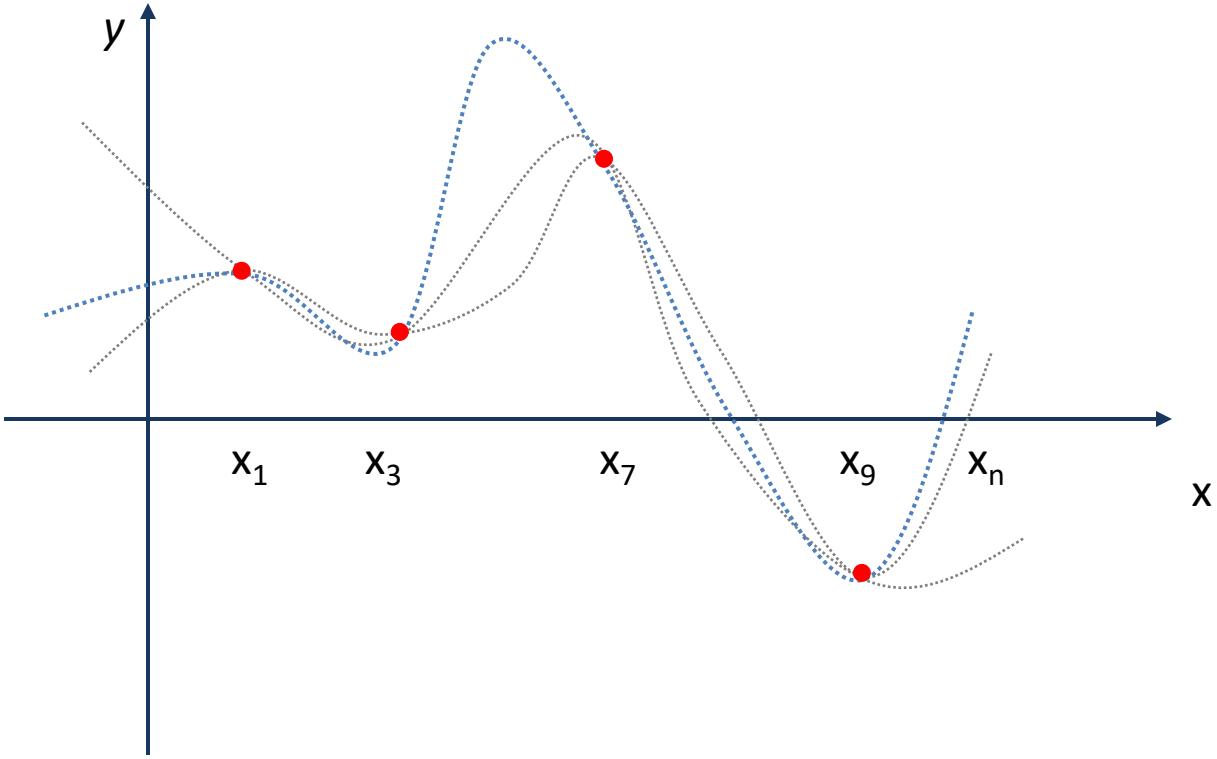
Let a function f: $\mathbf{x} \longrightarrow \mathbf{y}$ be in their **Joint Space**, such as ($\mathbf{x}$, $f(\mathbf{x})$) or ($\mathbf{x}$, $\mathbf{y}$).

Given a set of inputs $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, without prior knowledge, $f(\mathbf{x})$ can be as many as possible:



Function in Joint Space

Given a set of inputs $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, we now have a datasets of $(\mathbf{x}, f(\mathbf{x}))$ in their Joint Space, such that $\{ (\mathbf{x}_1,y_1), (\mathbf{x}_3,y_3), (\mathbf{x}_7,y_7), (\mathbf{x}_9,y_9)\}$.



As we can see, if we have enough data points, all possible dependencies of $(\mathbf{x}, f(\mathbf{x}))$ could be reduced.

Function in Joint Space

As we can see, if we don't have any prior knowledge on a function f: $\mathbf{x} \longrightarrow \mathbf{y}$, it is always better to estimate their dependency in their Joint Space ($\mathbf{x}$, $f(\mathbf{x})$), if we have enough data samples.

Actually, the **Process of Estimation** of the Functional Dependency in Joint Space (X, Y) is commonly known as **Gaussian Process**.

In **Gaussian Process**, instead of estimating a Deterministic **Output Function** $\mathbf{y} = f(\mathbf{x})$, we are estimating a Stochastic **Inference Function** $P(\mathbf{y} \mid \mathbf{x})$ in the Joint Space of $P(\mathbf{x}, \mathbf{y}) \subseteq (\mathbf{x} \times \mathbf{y})$.

Actually, we could generally define **Machine Learning** as the **Process of Estimation** of **Inference Function** $P(\mathbf{y} \mid \mathbf{x})$ in the Joint Space of $P(\mathbf{x}, \mathbf{y}) \subseteq (\mathbf{x} \times \mathbf{y})$.

Now, the question is how the Probabilities of $P(\mathbf{x})$ and $P(\mathbf{y})$ correlate in their Joint Space of $P(\mathbf{x}, \mathbf{y})$ to estimate **Inference Function** $P(\mathbf{y} \mid \mathbf{x})$.

Vector Space

Vector Space is a **Structured Set**, endowed with **Linear Structure**, which defines:

$$+ : V \times V \rightarrow V$$
$$. : \alpha \times V \rightarrow V$$

which implies:

$$+(u, v) \rightarrow u + v$$
$$.(a, u) \rightarrow au$$

Thus, a vector **v** in a Vector Space **V** will have a Linear Structure, which is **invariant** under **Linear Transformation**.

Vector Space

Actually, a Vector Space **V** is the **Linear Span** formed by the **Basis** { $\mathbf{g}_1$, $\mathbf{g}_2$, $\mathbf{g}_3$, … , $\mathbf{g}_n$ } such that:

$$\sum_{i=1}^{n} v_i \boldsymbol{g}_i = 0 \; iff \; v_i = 0$$

Any vector in the Vector Space can be uniquely identified by the Linear Combination of the Basis Vectors of the Vector Space.

Therefore, a **Vector** has two aspect: **Components** and **Basis**:

$$\boldsymbol{v} = \sum v_i \boldsymbol{g}_i$$

Probabilistic Vector

Now that we have defined Vectors in Vector Space, we will look into **Probabilistic Vectors** or **Stochastic Vectors**.

Although we might have been familiar with Probability, it would be uncommon to define **Probabilistic Vectors**.

As the vectors we know are usually represented by arrows, with magnitude and direction, it should naturally occurs to us as to how to understand probability as arrows.

However, **Information Geometry** has become essential as it would start to treat **Probabilities** as **Vectors** in **Topological Manifolds** to analyze their **Transformation** and **Connection**.

Probabilistic Vector

To understand what a Probabilistic Vector would look like, let us consider a following vector.

0.45	**0.55**
H	T

As we can see the Probability of Tossing a Coin could be considered as a Probabilistic Vector.

However, it would naturally occurs to us as to what would be the **Coordinates** of **Vectors** and **Covectors** of Stochastic Vector Space.

Probability Space

As we know already, a **Probability** is a **Possible Outcome** of a **Random Event**, such as *Flipping a Coin*, or *Throwing a Dice*.

Thus, a set of **All Possible Outcomes** of **Random Events**, along with their Probability, defines a **Probability Space**.

Probability Space is formally defined by a **Tuple**: (Ω, S, P).

- Ω represents the **Sample Space**, which is the **Space of All Possible Outcomes**
- S represents the **Space of Family of Events**, which is the **Power Set** of Ω
- P is defined as **Probability Measure**, which maps the **Events** to a **Set** [0, 1].

σ-Algebra

A family **S** of subsets of a sample space Ω is a **σ-algebra** defined by the following algebraic structures , such that **S** = { ∅, A, A_i, Ω }.

$$\Omega \in S$$

$$\bigcup_{i=1}^{\infty} A_i \in S$$

If **Ω** is a **topological space** the σ-algebra generated by all the open subset of **Ω** is called the **Borel's algebra**.

Therefore, we could assume that Probability Space **(Ω, S, P)** is a **Topological Space**, if such Probability Space has **σ-algebra**, which generates open subsets of **Ω**.

The space defined by **(Ω, S)** is known as **Measureable Space**.

Probability Measure

P is defined as **Probability Measure**, which maps the **Events** to a **Set** [0, 1].

As we already know, a **Random Events** is a Possible Outcome of an **Experiment** or **Trial**, such as Flipping a Coin: { H, T}:

$$\{\{\emptyset\}, \{H\}, \{T\}, \{H, T\}\} \rightarrow [0,1]$$

However, the problem is how do we define such a function *f* to represent **Probability Measure**?

Probability Measure

Formally, a measure **λ** on the measurable space (Ω,S) is a **map**, which maps events to real positive numbers such that:

$$\lambda: \mathbf{S} \rightarrow \mathrm{R}^{+}$$

$$\lambda[\emptyset] = 0$$

$$\lambda\left[\bigcup_{i=1}^{\infty} A_i\right] = \sum_{i=1}^{\infty} P[A_i]$$

A measure λ is said to be finite if $\lambda[\Omega] < \infty$.

Measurable Function

Let (E,B) a measurable space, a **measurable function** *f* from (Ω,S) to (E,B) is such the inverse image of a measurable set in B is a measurable set in S :

$$f^{-1}(B) = \{\omega \in \Omega, X(\omega) \in B\} \in S$$

Conversely,

$$f : \omega \to X(\omega), \omega \in \Omega, X(\omega) \in B$$

Such a measurable bijection function *f*:(Ω,S) ⟶(E,B) is known as **Random Variable**, represented by **X(ω)** or simply **X**.

Random Variable

A real **random variable** on the measurable space (Ω,S) is a measurable function **X** with value in $E \subseteq R$ (or $R = R \cup \{ -\infty, +\infty \}$) and B is the σ-algebra generated by the topology on E.

If E is a Vector Space, then **X(ω)** or **X** is known as **Random Vector**.

Now, from **Random Vectors**, we will look into **Probabilistic Vectors**.

Probabilistic Vector

As we already know a **Vector** has two aspect: **Components** and **Basis**, such that:

$$\boldsymbol{v} = \sum v_i \boldsymbol{g}_i$$

Similarly, we could represent **Probability** as **Probabilistic Vector**:

$$P(\boldsymbol{X}) = \sum \lambda_i(\omega_i)\boldsymbol{p}^i = \sum x_i \boldsymbol{p}^i$$

$$x_i = \lambda_i(\omega_i)$$

As we can see, $x_i = \lambda_i(\omega_i)$ is the **Random Variables** of Measurable Space (Ω,S), $\mathbf{p}^i$ is the **Basis** of **Probabilistic Vector Space**.

Coordinates of Probabilistic Vector

Actually, the **Basis** of **Probabilistic Vector Space** represents the **Coordinates** of **Probabilistic Vector Space**.

However, we should be aware that **Probabilistic Vector** is constrained by $\sum P(\mathbf{X}) = 1$.

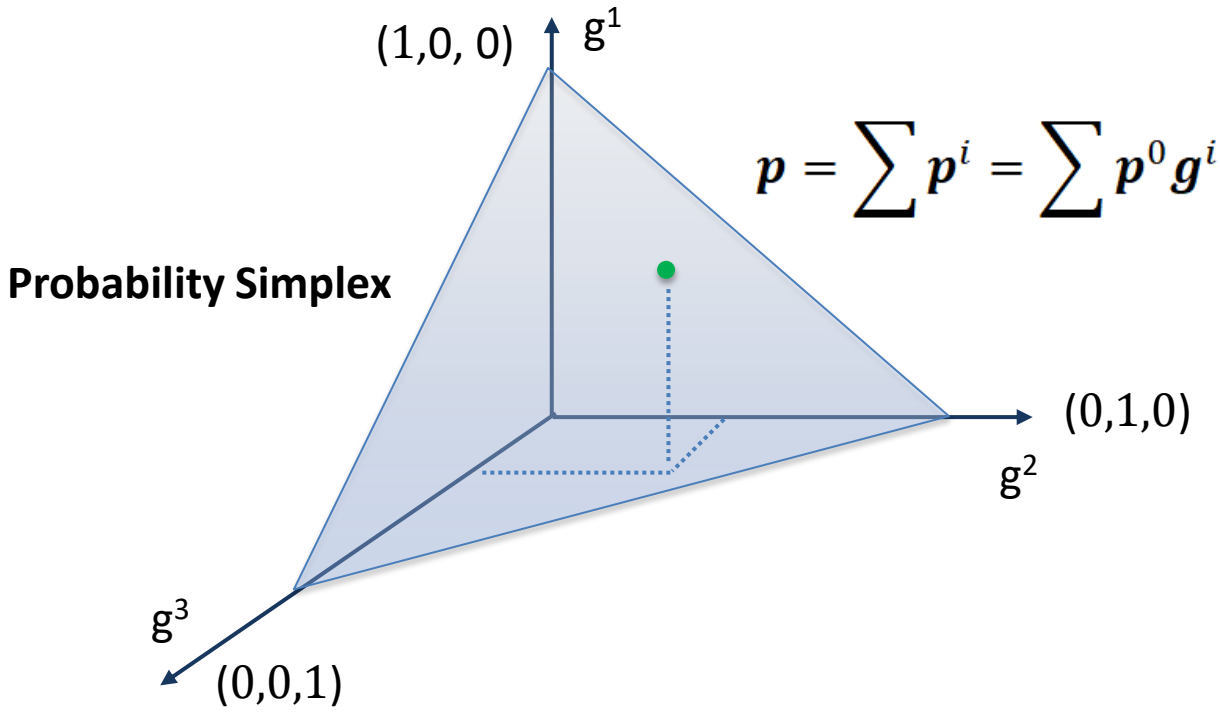
Thus, it should naturally occurs to us as to how **Coordinates** of **Probabilistic Vector Space** should look like.

One of the commonly used **Coordinates** of **Probabilistic Vector Space** is the **Barycentric Coordinates** defined in **Probability Simplex**.

Probability Simplex

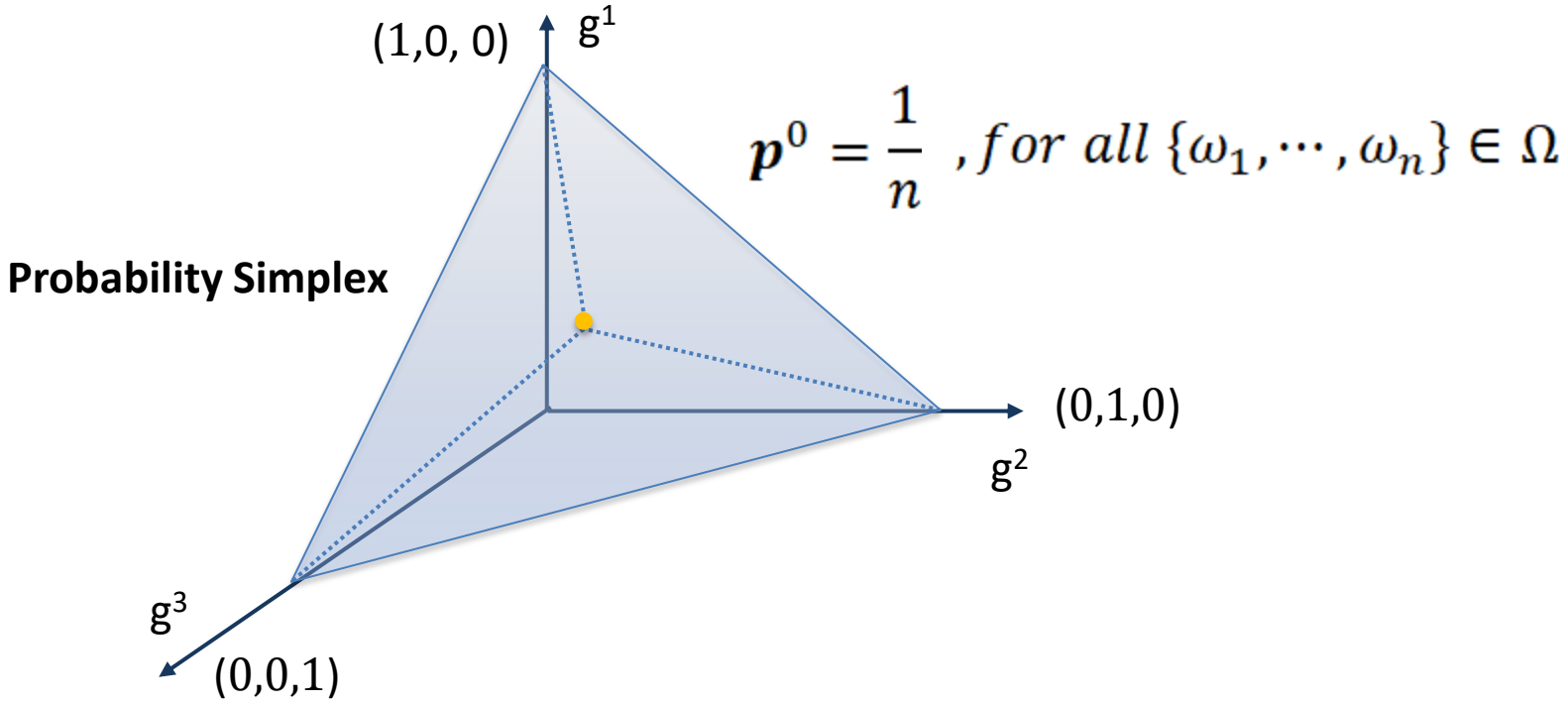
Suppose we have a **Measurable Space** (Ω,S), and the elements of Ω : { ω_1, …, ω_n }, and then we have random variables x_i : { $\lambda_1(\omega_1)$, …, $\lambda_n(\omega_n)$}.

Then, the **Probability Simplex** is defined by **Affine Coordinates** relative to the **Simplex Vertices** { $\mathbf{g}^1$,⋯, $\mathbf{g}^n$}, such that { (1, …, 0), …, (0, …, 1)}, where $\mathbf{p}^i = \mathbf{p}^0\mathbf{g}^i$.



Probability at Origin

Now, it should naturally occur to us as to what really is the Probability at Origin $\mathbf{p^0}$, which exists at the Center of Probability Simplex.



It turns out that $\mathbf{p^0}$ represents the **Equally Likelihood**, which represents the **Maximum Entropy**, such that $\mathbf{p^0}$ = 1/n for all the elements of Ω : { ω_1, …, ω_n }.

Mixture Coordinates

As we can see, **Barycentric Coordinates** can be defined as:

$$\boldsymbol{p} = \sum \boldsymbol{p}^i = \sum \boldsymbol{p}^0 \boldsymbol{g}^i$$

in which, each **Coordinate** of **Probability Simplex**, $\mathbf{p}^i$, can be considered as **Probability at Origin**, which is at the **Center of Probability Simplex**, in the Direction of $\mathbf{g}^i$.

Now, we can define **Mixture Coordinates**, which is represented by $\boldsymbol{\xi}^i = \mathbf{p}^i$.

Thus, **Mixture Coordinates** can be defined as:

$$\boldsymbol{\xi}^i = \boldsymbol{p}^i = \sum \boldsymbol{p}^0 \boldsymbol{g}^i$$

Mixture Coordinates

Therefore, with **Mixture Coordinates**, a **Probabilistic Vector** can be represented as:

$$P(\boldsymbol{X}) = \sum \lambda_i(\omega_i)\boldsymbol{p}^i = \sum x_i \boldsymbol{p}^i$$

$$= \sum x_i \boldsymbol{p}^0 \boldsymbol{g}^i$$

$$= \sum x_i \boldsymbol{\xi}^i$$

And of course, **Mixture Coordinates** are **Barycentric Coordinates** in **Probability Simplex**, which is measured from $\mathbf{p}^0$, which is at **Center of Probability Simplex**.

Exponential Coordinates

As we can see, **Barycentric Coordinates** is defined as:

$$\boldsymbol{p} = \sum \boldsymbol{p}^i = \sum \boldsymbol{p}^0 \boldsymbol{g}^i = \sum \boldsymbol{\xi}^i$$

Now, we are going to define another Coordinates of Probabilistic Vector, which is also known as **Exponential Coordinates**, which represented by θ^i, which is exponential.

$$\boldsymbol{p} = \sum \boldsymbol{p}^i = \sum \boldsymbol{p}^0 \exp\left(\boldsymbol{\theta}^i\right) = \sum \boldsymbol{p}^0 e^{\boldsymbol{\theta}^i} = \sum \boldsymbol{\xi}^i$$

Thus,

$$\boldsymbol{p}^0 \exp\left(\boldsymbol{\theta}^i\right) = \boldsymbol{p}^0 e^{\boldsymbol{\theta}^i} = \boldsymbol{\xi}^i = \boldsymbol{p}^i$$

Dual Coordinates

Now we can define the relationship between **Mixture Coordinates** ξ^i and **Exponential Coordinates** θ^i, which in fact are **Dual Coordinates** of Probabilistic Vector Space.

$$p^0 \exp(\theta^i) = p^0 e^{\theta^i} = \xi^i = p^i$$

$$\theta^i = log\left(\frac{p^i}{p^0}\right) = log\left(\frac{\xi^i}{p^0}\right)$$

Representation of Probabilistic Vector

In **Mixture Coordinates**, a Probabilistic Vector can be represented as:

$$P(\boldsymbol{X}) = \sum \lambda_i(\omega_i)\boldsymbol{p}^i = \sum x_i \boldsymbol{\xi}^i$$

But,

$$P(\boldsymbol{X}) = \sum x_i \boldsymbol{\xi}^i = \sum x_i \boldsymbol{p}^0 e^{\boldsymbol{\theta}^i}$$

$$\exp\left(\log\left(P(\boldsymbol{X})\right)\right) = exp\left(\sum log\left(x_i \boldsymbol{p}^0 e^{\boldsymbol{\theta}^i}\right)\right)$$

$$= exp\left(\sum \log(x_i)\,\boldsymbol{\theta}^i + \log\left(\boldsymbol{p}^0\right)\right)$$

$$P(\boldsymbol{X}) = exp\left(\sum x'_i \boldsymbol{\theta}^i + \log\left(\boldsymbol{p}^0\right)\right) \qquad \therefore x'_i = \log(x_i)$$

Representation of Probabilistic Vector

In **Mixture Coordinates**, a Probabilistic Vector can be represented as:

$$P(\boldsymbol{X}) = \sum \lambda_i(\omega_i)\boldsymbol{p}^i = \sum x_i \boldsymbol{\xi}^i$$

And in **Exponential Coordinates**,

$$P(\boldsymbol{X}) = exp\left(\sum x_i \boldsymbol{\theta}^i + \log\left(\boldsymbol{p}^0\right)\right)$$

And of course, for a probabilistic vector, like any other vector, any **Coordinate Transformation** should follow by the **Transformation of Components and Basis**.

Exponential Family of Probability

When we represent a Probabilistic Vector P(**X**) in Exponential Coordinates $\boldsymbol{\theta^i}$:

$$P(\boldsymbol{X}) = exp\left(\sum x_i \boldsymbol{\theta}^i + \log\left(\boldsymbol{p}^0\right)\right)$$

Actually, such representation of Probabilistic Vector P(**X**) in Exponential Coordinates **θ** results in **Generic Exponential Family of Probability**:

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = exp\left(\sum f_i(\boldsymbol{X})\, \boldsymbol{\theta}^i + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})\right) \quad \therefore x_i = f_i(\boldsymbol{X})$$

Statistical Manifold

Exponential Family of Probability is a Probabilistic Vector Space represented in Exponential Coordinates θ^i , parameterized by θ, given Random Variable **X**.

$$\boldsymbol{S} = \{ P(\boldsymbol{X} \mid \boldsymbol{\theta}) \}$$

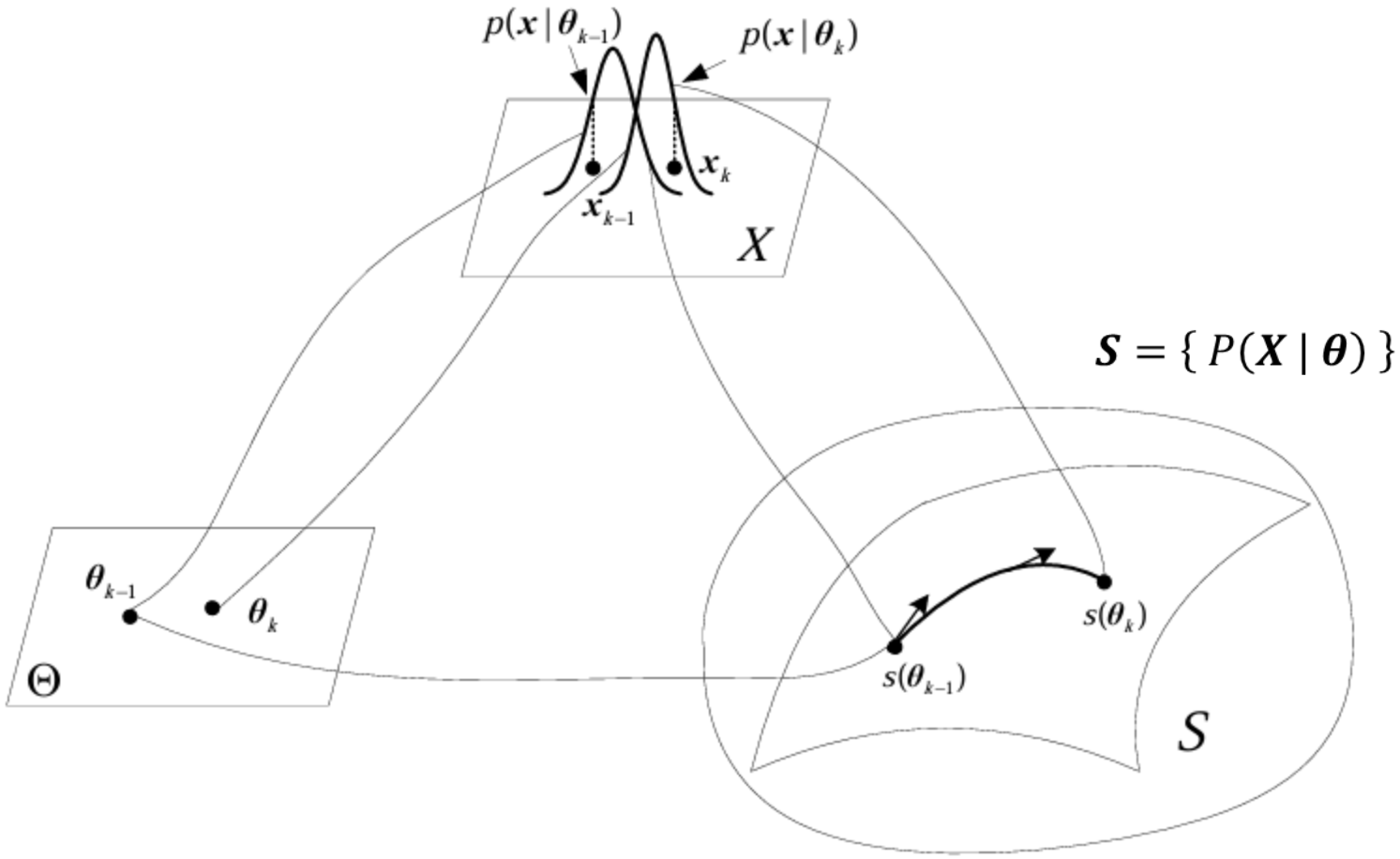
With **Exponential Family of Probability**, we could define **Statistical Manifold**, from the parameters θ, such that the **Topological Manifold** of **Probability Space** is Smooth, and thus **Differentiable**.

The parameterized family of probability distributions $\mathbf{S} = \{ P(\mathbf{X} \mid \boldsymbol{\theta})\}$, where **X** is a random variable and $\boldsymbol{\theta} = (\boldsymbol{\theta}^1,...,\boldsymbol{\theta}^n)$ is a parameter vector specifying the distribution.

The family **S** is regarded as a **Statistical Manifold** with $\boldsymbol{\theta}$ as its (possibly local) coordinate system.

Statistical Manifold

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = exp\left(\sum f_i(\boldsymbol{X})\boldsymbol{\theta}^i + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})\right)$$



Tangent Space of Statistical Manifold

$$\boldsymbol{S} = \{\, P(\boldsymbol{X} \mid \boldsymbol{\theta}) \,\}$$

Now, we will define the **Tangent Space** of **Statistical Manifold S** as the Gradient of **S**, $\nabla_{\theta}\,\mathbf{S}$. However, we are more interested in the Tangent Space represented by $\nabla_{\theta}\log(\mathbf{S})$, which in turn represented by $\nabla_{\theta}\log(P(\mathbf{X} \mid \theta))$.

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = exp\left(\sum f_i(\boldsymbol{X})\boldsymbol{\theta}^i + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})\right)$$

$$\nabla_{\theta} log(P(\boldsymbol{X} \mid \boldsymbol{\theta})) = \nabla_{\theta}\left(\sum f_i(\boldsymbol{X})\boldsymbol{\theta}^i + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})\right)$$

$$= \nabla_{\theta}(f(\boldsymbol{X})\boldsymbol{\theta} + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})) \;\; \therefore f(\boldsymbol{X})\boldsymbol{\theta} = \sum f_i(\boldsymbol{X})\boldsymbol{\theta}^i$$

$$= f(\boldsymbol{X})\nabla_{\theta}\boldsymbol{\theta} - \nabla_{\theta}\varphi(\boldsymbol{\theta})\nabla_{\theta}\boldsymbol{\theta}$$

$$= \nabla_{\theta}\boldsymbol{\theta}(f(\boldsymbol{X}) - \nabla_{\theta}\varphi(\boldsymbol{\theta}))$$

Tangent Space of Statistical Manifold

Now, the Tangent Space represented by $\nabla_{\theta}\log(\mathbf{S})$ is defined as:

$$\nabla_{\theta} log(\boldsymbol{S}) = \{ \nabla_{\theta} \boldsymbol{\theta} (f(\boldsymbol{X}) - \nabla_{\theta} \varphi(\boldsymbol{\theta})) \}$$

However, what is particularly interesting is $\nabla_{\boldsymbol{\theta}}\varphi(\boldsymbol{\theta})$. Now let us look into what really is $\varphi(\boldsymbol{\theta})$.

As we know,

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = exp\left(\sum f_i(\boldsymbol{X})\boldsymbol{\theta}^i + g(\boldsymbol{X}) - \varphi(\boldsymbol{\theta})\right)$$

Thus,

$$\int P(\boldsymbol{X} \mid \boldsymbol{\theta}) d\mu(\boldsymbol{X}) = 1 \qquad \therefore d\mu(\boldsymbol{X}) = \exp\big(g(\boldsymbol{X})\big)\, d\boldsymbol{X}$$

Free Energy Function

Thus,

$$\int exp(f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta}))d\mu(\boldsymbol{X}) = 1$$

Since exp(-$\varphi(\boldsymbol{\theta})$) does not depend on X, we can pull it out of the Integral.

$$exp\big(\varphi(\boldsymbol{\theta})\big) = \int exp(f(\boldsymbol{X})\boldsymbol{\theta})d\mu(\boldsymbol{X})$$

$$\varphi(\boldsymbol{\theta}) = log\left(\int exp(f(\boldsymbol{X})\boldsymbol{\theta})d\mu(\boldsymbol{X})\right)$$

Now, $\varphi(\boldsymbol{\theta})$ is defined as **Free Energy Function**, which is also known as **Potential Function** of $\boldsymbol{\theta}$.

Gradient of Free Energy Function

Now we will continue to look in $\nabla_{\boldsymbol{\theta}}\varphi(\boldsymbol{\theta})$.

$$exp\big(\varphi(\boldsymbol{\theta})\big) = \int exp(f(\boldsymbol{X})\boldsymbol{\theta})d\mu(\boldsymbol{X})$$

$$\frac{\partial}{\partial\boldsymbol{\theta}} exp\big(\varphi(\boldsymbol{\theta})\big) = \int \frac{\partial}{\partial\boldsymbol{\theta}} exp(f(\boldsymbol{X})\boldsymbol{\theta})d\mu(\boldsymbol{X})$$

$$exp\big(\varphi(\boldsymbol{\theta})\big)\frac{\partial\varphi(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} = \int f(\boldsymbol{X})exp(f(\boldsymbol{X})\boldsymbol{\theta})d\mu(\boldsymbol{X})$$

$$\frac{\partial\varphi(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} = \int f(\boldsymbol{X})exp\big(f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta})\big)d\mu(\boldsymbol{X})$$

$$= \int f(\boldsymbol{X})P(\boldsymbol{X}\mid\boldsymbol{\theta})d\mu(\boldsymbol{X})$$

Gradient of Free Energy Function

As we can see, the **Gradient** of **Free Energy Function** is the **Expected Value** of $f(\mathbf{X})$.

$$\frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \int f(\boldsymbol{X}) P(\boldsymbol{X} \mid \boldsymbol{\theta}) d\mu(\boldsymbol{X})$$

$$\frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = E[f(\boldsymbol{X})]$$

Now, we can redefine the **Tangent Space** of $\nabla_\theta \log(\mathbf{S})$ as:

$$\nabla_\theta log(\boldsymbol{S}) = \{ \nabla_\theta \boldsymbol{\theta} \underbrace{(f(\boldsymbol{X}) - E[f(\boldsymbol{X})])}_{X - \bar{X}} \}$$

Transformation and Connection

Like any other Topological Manifold **M**, Statistical Manifold **S** also need to define **Transformation** and **Connection** as well.

First, each **Tangent Space T_p** exists differently from the Manifold **S** represented by Local Coordinates θ, with different bases. Therefore, we always have to define **Coordinate Transformation**.

$$g_t : \boldsymbol{\theta} \rightarrow T_p \boldsymbol{\theta}$$

Second, each **Tangent Space T_p** exists differently from other **Tangent Space T_q** in the **Tangent Bundles** of Statistical Manifolds. Therefore, we also need to define **Connections** between **Tangent Spaces** in the **Tangent Bundles** of Statistical Manifolds.

$$g_c : T_p \boldsymbol{\theta} \times T_q \boldsymbol{\theta} \rightarrow R$$

Convex Function

The **Free Energy Function** $\varphi(\boldsymbol{\theta})$ can be considered as a **Smooth Convex Function** $\varphi(\boldsymbol{\theta})$ defined in an open set ***S*** of R^n, where **θ** plays the role of a coordinate system.

As we can see, the basis of the Tangent Space T_p of the Local Coordinate **θ** { $\boldsymbol{\theta_1}$, …, $\boldsymbol{\theta_n}$} is $\partial_{\boldsymbol{\theta}}$ { $\partial_{\boldsymbol{\theta 1}}$, …, $\partial_{\boldsymbol{\theta n}}$ }.

Thus, the **Metric Tensor** g_{ij} of Statistical Manifold **S** is defined as:

$$g_{ij} = \frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_i} \cdot \frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_j}$$

$$= \frac{\partial \varphi^2(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}$$

Metric Tensor

As we can see, if the **Metric Tensor** g_{ij} of Statistical Manifold **S** is a Diagonal Matrix, such that

$$\frac{\partial \varphi^2}{\partial \theta_i \partial \theta_j} = \frac{\partial \varphi^2}{\partial \theta_j \partial \theta_i}$$

Then, such a Metric Tensor is defined as **Riemannian Metric**, and a Manifold, including Statistical Manifold, endowed with Riemannian Metric, is called **Riemannian Manifold**.

In such a special case, $\varphi(\boldsymbol{\theta})$, however, would have been reduced as:

$$\varphi(\boldsymbol{\theta}) = \frac{1}{2}\sum(\boldsymbol{\theta}_i)^2$$

Legendre Transformation

As the **Free Energy Function** $\varphi(\boldsymbol{\theta})$ can be considered as a **Smooth Convex Function**, we can transform **θ** to **η** by Legendre Transformation, such that:

$$\eta^j = \frac{\partial}{\partial \theta_i} \varphi(\boldsymbol{\theta})$$

Thus, we can define another Convex Function $\varphi^*(\boldsymbol{\eta})$, which is dual to $\varphi(\boldsymbol{\theta})$ such that:

$$\begin{aligned} \varphi^*(\boldsymbol{\eta}) &= \boldsymbol{\theta} \cdot \boldsymbol{\eta} - \varphi(\boldsymbol{\theta}) \\ &= \theta_i \eta^j - \varphi(\boldsymbol{\theta}) \end{aligned}$$

From this equation, we can conclude that:

$$\theta_i = \frac{\partial}{\partial \eta^i} \varphi^*(\boldsymbol{\eta})$$

Dual Coordinates

Now, we can also define a **Metric Tensor** g^{ij} in Dual Coordinates of **η** such that:

$$g^{ij} = \frac{\partial \varphi^*(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^i} \cdot \frac{\partial \varphi^*(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^i}$$

$$g_{ij} = \frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_i} \cdot \frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_j}$$

$$g^{ij} = \frac{\partial \theta_{\mathrm{i}}}{\partial \eta^i} \qquad g_{ij} = \frac{\partial \eta^i}{\partial \theta_{\mathrm{i}}}$$

Expectation Coordinates

After defining Dual Coordinates of **η**, it should naturally occurs to us as to what is such Dual Coordinates **η**.

Interestingly, Dual Coordinates **η** is known as **Expectation Coordinates**, which actually measures the **Expectation** of **Probability**, such that:

$$\boldsymbol{\eta} = \frac{\partial \varphi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = E\big(f(\boldsymbol{X})\big)$$

Then, we can redefine the **Tangent Space** of $\nabla_{\theta} \log(\boldsymbol{S})$ as:

$$\nabla_{\theta} log(\boldsymbol{S}) = \{\, \nabla_{\theta} \boldsymbol{\theta} (f(\boldsymbol{X}) - \boldsymbol{\eta}) \,\}$$

Negative Entropy

Entropy is generally defined as the **Expectation of Uncertainty** such that:

$$H(\boldsymbol{x}) = -E[log(P(\boldsymbol{x}))] = -\int P(\boldsymbol{x}) log(P(\boldsymbol{x})) d\boldsymbol{x}$$

Now, we will look into how Dual Coordinates **η** is related to **Negative Entropy**, –H(X).

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = exp(f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta}))$$

$$H(P(\boldsymbol{X} \mid \boldsymbol{\theta})) = -E[log(P((\boldsymbol{X} \mid \boldsymbol{\theta}))]$$

$$= -E[f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta})]$$

$$= -\boldsymbol{\theta} E[f(\boldsymbol{X})] + \varphi(\boldsymbol{\theta})$$

$$= -\boldsymbol{\theta}\boldsymbol{\eta} + \varphi(\boldsymbol{\theta}) = -\varphi^*(\boldsymbol{\eta})$$

Dual Convex Functions

Free Energy Function $\varphi(\boldsymbol{\theta})$ and **Negative Entropy Function** $\varphi^*(\boldsymbol{\eta})$ are actually dual convex functions of each other.

$$\varphi(\boldsymbol{\theta}) = log \int exp(\boldsymbol{\theta} \cdot \boldsymbol{X}) d\mu(\boldsymbol{X})$$

$$\varphi^*(\boldsymbol{\eta}) = \int P(\boldsymbol{X} \mid \boldsymbol{\eta}) log(P(\boldsymbol{X} \mid \boldsymbol{\eta})) d\boldsymbol{X}$$

And $\varphi(\boldsymbol{\theta})$ and $\varphi^*(\boldsymbol{\eta})$ are related by:

$$\varphi^*(\boldsymbol{\eta}) = \boldsymbol{\theta} \cdot \boldsymbol{\eta} - \varphi(\boldsymbol{\theta})$$

Score Function

We can define $\nabla_\theta \log(\mathbf{S})$, **Tangent Vector** of Statistical Manifold **S**, as **Score Function**.

$$\nabla_\theta log(\boldsymbol{S}) = \nabla_\theta log(P(X|\theta)) = \frac{\nabla_\theta P(X|\theta)}{P(X|\theta)}$$

Thus, the Expectation of Score Function $E[\nabla_\theta \log(\mathbf{S})]$ is:

$$E[\nabla_\theta log(\boldsymbol{S})] = \int \frac{\nabla_\theta P(X|\theta)}{P(X|\theta)} P(X|\theta) dX$$

$$= \nabla_\theta \int P(X|\theta) dX = \nabla_\theta(1) = 0$$

Hessian of Score Function

Now, let us look into the **Hessian of the Score Function**:

$$\nabla_\theta{}^2 log(\boldsymbol{S}) = \nabla_\theta \left[\frac{\nabla_\theta P(X|\theta)}{P(X|\theta)} \right]$$

$$= \frac{\nabla_\theta{}^2 P(X|\theta)}{P(X|\theta)} - \frac{\nabla_\theta P(X|\theta)\, \nabla_\theta P(X|\theta)}{P(X|\theta)^2}$$

$$= \frac{\nabla_\theta{}^2 P(X|\theta)}{P(X|\theta)} - \nabla_\theta log(\boldsymbol{S}) \nabla_\theta log(\boldsymbol{S})$$

Expectation of Hessian of Score Function

Now, let us defined the **Expectation of the Hessian of the Score Function**:

$$E[\nabla_\theta{}^2 log(\boldsymbol{S})] = \int \left[\frac{\nabla_\theta{}^2 P(X|\theta)}{P(X|\theta)} - \nabla_\theta log(\boldsymbol{S}) \nabla_\theta log(\boldsymbol{S}) \right] P(X|\theta) dX$$

$$= \int \nabla_\theta{}^2 P(X|\theta) dX - \int \nabla_\theta log(\boldsymbol{S}) \nabla_\theta log(\boldsymbol{S}) P(X|\theta) dX$$

$$= 0 - E[\nabla_\theta log(\boldsymbol{S}) \nabla_\theta log(\boldsymbol{S})] \quad \therefore E[\nabla_\theta log(\boldsymbol{S})] = 0$$

$$= -E[(\nabla_\theta log(\boldsymbol{S}) - E[\nabla_\theta log(\boldsymbol{S})])^2]$$

$$= -Var[\nabla_\theta log(\boldsymbol{S})]$$

Fisher Information Metric

Now, let us defined the **Variance of the Score Function**, as **Fisher Information Metric** I(θ):

$$E[\nabla_\theta{}^2 log(\boldsymbol{S})] = -Var[\nabla_\theta log(\boldsymbol{S})] = -I(\theta)$$

Thus, **Fisher Information Metric** is the **Variance of Score Function** or **Negative Expectation of Hessian of Score Function**.

$$I(\theta) = Var[\nabla_\theta log(\boldsymbol{S})] = -E[\nabla_\theta log(\boldsymbol{S}) \nabla_\theta log(\boldsymbol{S})] = -E\left[\nabla_\theta{}^2 log(\boldsymbol{S})\right]$$

Energy Function

As we know already, the Probabilistic Vector is defined as:

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = \exp\left(f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta})\right)$$

However, we could define f(**X**)θ or **F(X)**.θ as a function, which is commonly known as **Energy Function** $\mathcal{E}(\mathbf{X}, \theta)$, such that:

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) = -f(\boldsymbol{X})\boldsymbol{\theta}$$

Now let us define a **Partition Function Z** , from **Energy Function** $\mathcal{E}(\mathbf{X}, \theta)$, such that:

$$Z(\boldsymbol{X}, \boldsymbol{\theta}) = \int exp\big(-\varepsilon(\boldsymbol{X}, \boldsymbol{\theta})\big) d\mu(\boldsymbol{X})$$

Energy Function

Now, Probability P (**X** |**θ**) can be redefined as:

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = \frac{\exp\left(-\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\right)}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})}$$

However, as we know:

$$\varphi(\boldsymbol{\theta}) = -log\left(z(\boldsymbol{X}, \boldsymbol{\theta})\right)$$

Thus,

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = \exp\left(\varphi(\boldsymbol{\theta}) - \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\right)$$

Now, **Probability** can be defined as **Exponential of the Difference** between **Free Energy** and **Energy Function**.

Expectation of Energy Function

Now, the **Expectation of Energy Function** $E[\mathcal{E}(\mathbf{X}, \theta)]$ can be defined as:

$$E[\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})] = \int \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) P(\boldsymbol{X} \mid \boldsymbol{\theta}) dX$$

Thus:

$$E[\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})] = \int \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) \frac{\exp\big(-\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\big)}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})} dX$$

$$= \frac{1}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})} \int \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) \exp\big(-\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\big) dX$$

We can actually define $E[\mathcal{E}(\mathbf{X}, \theta)]$ as the **Gradient of Free Energy Function** $\nabla_{\theta} \varphi(\theta)$.

Expectation of Energy Function

However, as we know:

$$\varphi(\boldsymbol{\theta}) = -log\big(z(\boldsymbol{X}, \boldsymbol{\theta})\big)$$

Thus,

$$\frac{\partial}{\partial \boldsymbol{\theta}}\varphi(\boldsymbol{\theta}) = -\frac{1}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})}\frac{\partial}{\partial \boldsymbol{\theta}}\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})$$

$$= -\frac{1}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})}\int \frac{\partial}{\partial \boldsymbol{\theta}}\exp\big(-\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\big)dX$$

$$= \frac{1}{\boldsymbol{Z}(\boldsymbol{X}, \boldsymbol{\theta})}\int \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\exp\big(-\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})\big)dX$$

$$= E[\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})]$$

Variance of Energy Function

Since

$$log(P(\boldsymbol{X} \mid \boldsymbol{\theta})) = \varphi(\boldsymbol{\theta}) - \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})$$

$$\frac{\partial}{\partial \boldsymbol{\theta}} log(P(\boldsymbol{X} \mid \boldsymbol{\theta})) = \frac{\partial}{\partial \boldsymbol{\theta}} \varphi(\boldsymbol{\theta}) - \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})$$

If we substitute $\nabla_{\theta} \log(P(\mathbf{X} \mid \theta))$ in **Fisher Information Metric**:

$$I(\boldsymbol{\theta}) = \int \left(\frac{\partial}{\partial \boldsymbol{\theta}} \varphi(\boldsymbol{\theta}) - \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) \right)^2 P(\boldsymbol{X} \mid \boldsymbol{\theta}) d\boldsymbol{X}$$

$$= \int \left(E[\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})] - \mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta}) \right)^2 P(\boldsymbol{X} \mid \boldsymbol{\theta}) d\boldsymbol{X}$$

$$= Var[\mathcal{E}(\boldsymbol{X}, \boldsymbol{\theta})]$$

Hessian of Free Energy Function

However, as we know:

$$P(\boldsymbol{X} \mid \boldsymbol{\theta}) = \exp\left(f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta})\right)$$

Thus,

$$log\big(P(\boldsymbol{X} \mid \boldsymbol{\theta})\big) = f(\boldsymbol{X})\boldsymbol{\theta} - \varphi(\boldsymbol{\theta})$$

$$\frac{\partial}{\partial \boldsymbol{\theta}} log\big(P(\boldsymbol{X} \mid \boldsymbol{\theta})\big) = \frac{\partial}{\partial \boldsymbol{\theta}} f(\boldsymbol{X})\boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \varphi(\boldsymbol{\theta})$$

$$= f(\boldsymbol{X}) - \frac{\partial}{\partial \boldsymbol{\theta}} \varphi(\boldsymbol{\theta})$$

Hessian of Free Energy Function

$$\frac{\partial^2}{\partial\theta_i\partial\theta_j} log\big(P(\boldsymbol{X}\mid\boldsymbol{\theta})\big) = \frac{\partial}{\partial\boldsymbol{\theta}}\left(f(\boldsymbol{X}) - \frac{\partial}{\partial\boldsymbol{\theta}}\varphi(\boldsymbol{\theta})\right)$$

$$= -\frac{\partial^2}{\partial\theta_i\partial\theta_j}\varphi(\boldsymbol{\theta})$$

$$= -g_{ij}$$

But, $$I(\boldsymbol{\theta}) = E\left[\frac{\partial^2}{\partial\theta_i\partial\theta_j} log\big(P(\boldsymbol{X}\mid\boldsymbol{\theta})\big)\right]$$

$$= E\left[-\frac{\partial^2}{\partial\theta_i\partial\theta_j}\varphi(\boldsymbol{\theta})\right] = \frac{\partial^2}{\partial\theta_i\partial\theta_j}\varphi(\boldsymbol{\theta}) = g_{ij}$$

Divergence

In a **Dually Flat** Statistical Manifold **S**, a **Canonical Divergence** between Coordinates of $\boldsymbol{\theta}_P$ and $\boldsymbol{\theta}_Q$ is uniquely determined.

$$D[\boldsymbol{\theta}_P : \boldsymbol{\theta}_Q] = \varphi(\boldsymbol{\theta}_P) + \varphi^*(\boldsymbol{\eta}_Q) - \boldsymbol{\theta}^i{}_P \boldsymbol{\eta}_{Q^j}$$

$$D^*[\boldsymbol{\theta}_P : \boldsymbol{\theta}_Q] = \varphi^*(\boldsymbol{\eta}_P) + \varphi(\boldsymbol{\theta}_Q) - \boldsymbol{\eta}^i{}_P \boldsymbol{\theta}_{Q^j}$$

As we can see, Divergence is **dual**, but **asymmetrical**.

$$D[\boldsymbol{\theta}_P : \boldsymbol{\theta}_P] = 0$$

$$D[\boldsymbol{\theta}_P : \boldsymbol{\theta}_Q] \neq D[\boldsymbol{\theta}_Q : \boldsymbol{\theta}_P]$$

KL Divergence

Kullback–Leibler Divergence, $D_{KL}[P(\mathbf{X} \mid \theta_P) \;||\; Q(\mathbf{X} \mid \theta_P)]$ which is also known as **Relative Entropy**, between $P(\mathbf{X} \mid \theta_P)$ and $Q(\mathbf{X} \mid \theta_Q)$ is defined as:

$$D_{KL}[P(\boldsymbol{X} \mid \boldsymbol{\theta_P}) \,||Q(\boldsymbol{X} \mid \boldsymbol{\theta_Q})] = \int P(\boldsymbol{X} \mid \boldsymbol{\theta_P}) log\left(\frac{P(\boldsymbol{X} \mid \boldsymbol{\theta_P})}{Q(\boldsymbol{X} \mid \boldsymbol{\theta_Q})}\right) d\boldsymbol{X}$$

$$= E\left[log\left(\frac{P(\boldsymbol{X} \mid \boldsymbol{\theta_P})}{Q(\boldsymbol{X} \mid \boldsymbol{\theta_Q})}\right)\right]$$

As we can see, from KL Divergence, we could derive both **Metric** and **Connection** between two Statistical Manifold **P** and **Q**.

Sanov's Theorem

Sanov's Theorem actually tell us how unlikely it is for a random sample to exhibit a probability distribution that is different from the true underlying distribution.

It can be summarized as the **Probability of Unusual Data** from **Empirical Distribution** is inversely proportional to the **Exponential Function** of **Sample Size** and the **Divergence between True and Unusual Distribution**.

$$\boldsymbol{P}[P_n \approx Q] = e^{-nD(P \,||\, Q)}$$

More precisely,

$$\lim_{n\to\infty} \frac{1}{n} log(P_n \in Q) = -inf[D(P \,||\, Q)]$$

where, P is **True Distribution**, Q is a **Set of Unusual Distributions**, P_n is the **Empirical Distribution**, and *inf* is the **Infimum**, which represents the **greatest lower bound**.

Divergence and Distance

As we can see, Divergence can measure the **Difference** between Probability Distributions.

However, **Divergence** is not **Distance Measure**, though it is a **Difference Measure**, and thus, it is not **Communicative**.

$$D[\boldsymbol{\theta}_P : \boldsymbol{\theta}_Q] \neq D[\boldsymbol{\theta}_Q : \boldsymbol{\theta}_P]$$

As we know, **Fisher Information Metric** is a **Distance Measure**, and we will look into how Fisher Information Metric and Divergence are related.

Simply, it is because **Distance Measure** is far more useful than **Difference Measure**.

Divergence and Distance

Let us consider two nearby Probability Distributions P(X | θ) and P(X | θ + d θ)

Then their **Divergence D_{KL}** can be defined as:

$$D_{KL} = D_{KL}[P(X \mid \theta) \,||\, P(X \mid \theta + d\theta)]$$

Thus,

$$D_{KL} = \int P(X \mid \theta) log\left(\frac{P(X \mid \theta)}{P(X \mid \theta + d\theta)}\right) d\boldsymbol{X}$$

Divergence and Distance

We can define a L(X | θ) as:

$$L(X \mid \theta) = log\big(P(X \mid \theta)\big)$$

Then, by Taylor Series, L(X | θ + dθ) can be defined as:

$$L(X \mid \theta + d\theta) = L(X \mid \theta) + \frac{\partial L}{\partial \theta_i} d\theta^i + \frac{1}{2} \frac{\partial L^2}{\partial \theta_i \partial \theta_j} d\theta^i d\theta^j + O(\|d\theta^3\|)$$

Now, by substituting L(X | θ) and L(X | θ + dθ) :

$$D_{KL} = -E_\theta [L(X \mid \theta + d\theta) - L(X \mid \theta)]$$

Divergence and Distance

Therefore:

$$D_{KL} = -E_\theta \left[\frac{\partial L}{\partial \theta_i}\right] d\theta^i - \frac{1}{2} E_\theta \left[\frac{\partial L^2}{\partial \theta_i \partial \theta_j}\right] d\theta^i d\theta^j + O(\|d\theta^3\|)$$

As we know,

$$E_\theta \left[\frac{\partial L}{\partial \theta_i}\right] = 0 \quad E_\theta \left[\frac{\partial L^2}{\partial \theta_i \partial \theta_j}\right] = I_\theta$$

Thus,

$$D_{KL} = -\frac{1}{2} I_\theta d\theta^i d\theta^j + O(\|d\theta^3\|)$$